TriAlign

An agentic tool for clinically meaningful comparison of trial eligibility criteria

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Clinical protocols $\rightarrow$ reviewed structure $\rightarrow$ interpretable comparison

Why this tool is needed

Eligibility criteria determine who can enter a trial, yet important differences are buried in long protocols and inconsistent prose.

- Study teams compare criteria for protocol design, feasibility, competitive intelligence, evidence synthesis, and amendment review.
- Keyword search finds shared words; ordinary embeddings often find shared topics. Neither guarantees shared eligibility meaning.
- Users need to know whether criteria are equivalent, narrower, overlapping, contradictory, related, or incomparable—with evidence and uncertainty.

Practical goal

Select a criterion, retrieve comparable criteria across protocols, inspect the differences, and trace the result back to its source.

From AutoCriteria to Modern AutoCrit

AutoCriteria established

- LLM-assisted conversion of eligibility prose into structured criteria
- a scalable starting point for protocol preprocessing

Modern AutoCrit adds

- NCT, pasted-text, and full-protocol PDF ingestion
- atomic thresholds, negation, timing, repetition, exceptions, and AND/OR scope
- provenance, stable IDs, validation, human correction, and resumable jobs
- versioned Excel and validated JSON exports

Engineering change

The OpenAI SDK and small testable modules replaced LangChain/LangGraph and the original long script, making state, retries, validation, and provenance explicit.

Datta et al., AutoCriteria (2023); github.com/JonathanMa03/modern-autocrit

The gap TriAlign addresses

Pair	Embedding problem	Clinical relation
Age ≥ 18 vs. age < 18	nearly identical wording	disjoint populations
No cardiovascular disease vs. controlled hypertension permitted	same clinical topic	exception changes scope
No prior chemotherapy vs. none within 6 months	high lexical overlap	temporal rule differs
BP $> 145/95$ once vs. twice in 3 months	same measurement	repetition/timing differ

Three questions

Where do conventional embeddings fail? Can reviewed hard cases improve a dense encoder? Can an agent choose the right tools, quantify uncertainty, and escalate safely?

Protocol corpus: two complementary areas

50 oncology trials

Biomarkers, stage, histology, prior therapy, washout periods, performance status, organ function, and nested exceptions.

50 cardiovascular trials

Heart failure, atrial fibrillation, coronary/valvular disease, hypertension, stroke, thrombosis, PAD, and aneurysm.

- Target: approximately 2,000 human-reviewed atomic criteria.
- Chia may supplement terminology and examples, but does not supply gold pairwise labels.
- Every atom retains NCT ID, source version, parent statement, context, review, and validation.

Kury et al., Chia, Scientific Data 7, 281 (2020).

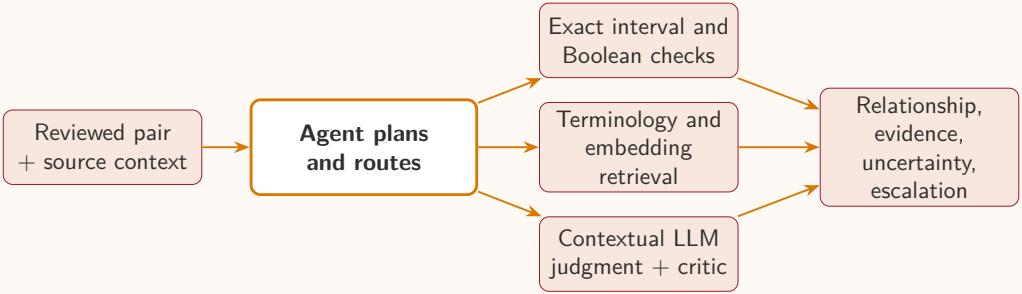
Interpretable relationships, not an opaque score

- **Equivalent** same operational rule
- **More restrictive** one permits a subset
- **Partially overlapping** shared but non-nested
- **Contradictory / disjoint** cannot both hold
- **Related, not comparable** shared topic, different operation
- **Unrelated** no meaningful relationship
- **Ambiguous** insufficient context

Why this matters

Similarity can rank candidates, but users see a clinical relationship, evidence, and calibrated uncertainty rather than a weighted mixture with unclear meaning.

The agentic workflow takes center stage



The agent selects tools, reconciles disagreements, records its trace, and abstains when evidence is insufficient. Exact cases are not forced through an LLM; ambiguous cases are not hidden behind cosine similarity.

Human steering becomes representation learning

1. The agent retrieves pairs and assembles structural and source evidence.
2. An LLM proposes a relationship; exact checks or a critic challenge conflicts.
3. A human accepts, corrects, or marks ambiguity. The human decision is authoritative.
4. Reviewed pairs form positives, hard negatives, and ordered anchors.
5. One open dense encoder is fine-tuned offline and calibrated on held-out data.
6. Later review prioritizes uncertainty and disagreement, compared with random sampling.

Contribution

A reviewed protocol dataset, clinically difficult negatives, agentic labeling workflow, and controlled evaluation—not merely checkpoint fine-tuning.

Evaluation plan

Component	Measures
Retrieval	precision/recall@k, mean reciprocal rank, anchor-ordering accuracy
Relationship prediction	macro-F1, class errors, confusion matrix, order consistency
Uncertainty	calibration, risk–coverage, abstention and escalation errors
Agent usefulness	correction rate, conflict detection, reviewer time, latency, and cost
Statistics	trial-family bootstrap intervals, paired comparisons, and ablations

Trial families are split before pair construction; amendments and shared templates stay together. Held-out labels remain unavailable to agent retrieval.

What we will build and how

Working tool

- cross-protocol criterion retrieval
- field-level and contextual differences
- relationship and uncertainty
- traceable source evidence
- correction and escalation

Execution

1. calibrate on 10 trials
2. extract/review the 100-trial cohort
3. measure embedding failures and label pairs
4. fine-tune and calibrate one encoder
5. integrate agent, critic, audit, and abstention

Safeguards

Human-reviewed inputs, blinded test labels, trial-family splits, and no automated patient screening, causal claims, or opaque hybrid score.

Take-home proposal

TriAlign

An agentic clinical data-science tool combining reviewed protocol structure, exact checks, retrieval, embeddings, contextual LLM reasoning, and human steering to compare eligibility criteria with interpretable relationships and calibrated uncertainty.

Modern AutoCrit supplies traceable atoms.

Human review supplies trustworthy relationships.

TriAlign learns and operationalizes the comparison.

References: Datta et al., AutoCriteria (2023); Kury et al., Chia (2020); Modern AutoCrit repository; full methods in the project proposal.